



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ANAIYAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

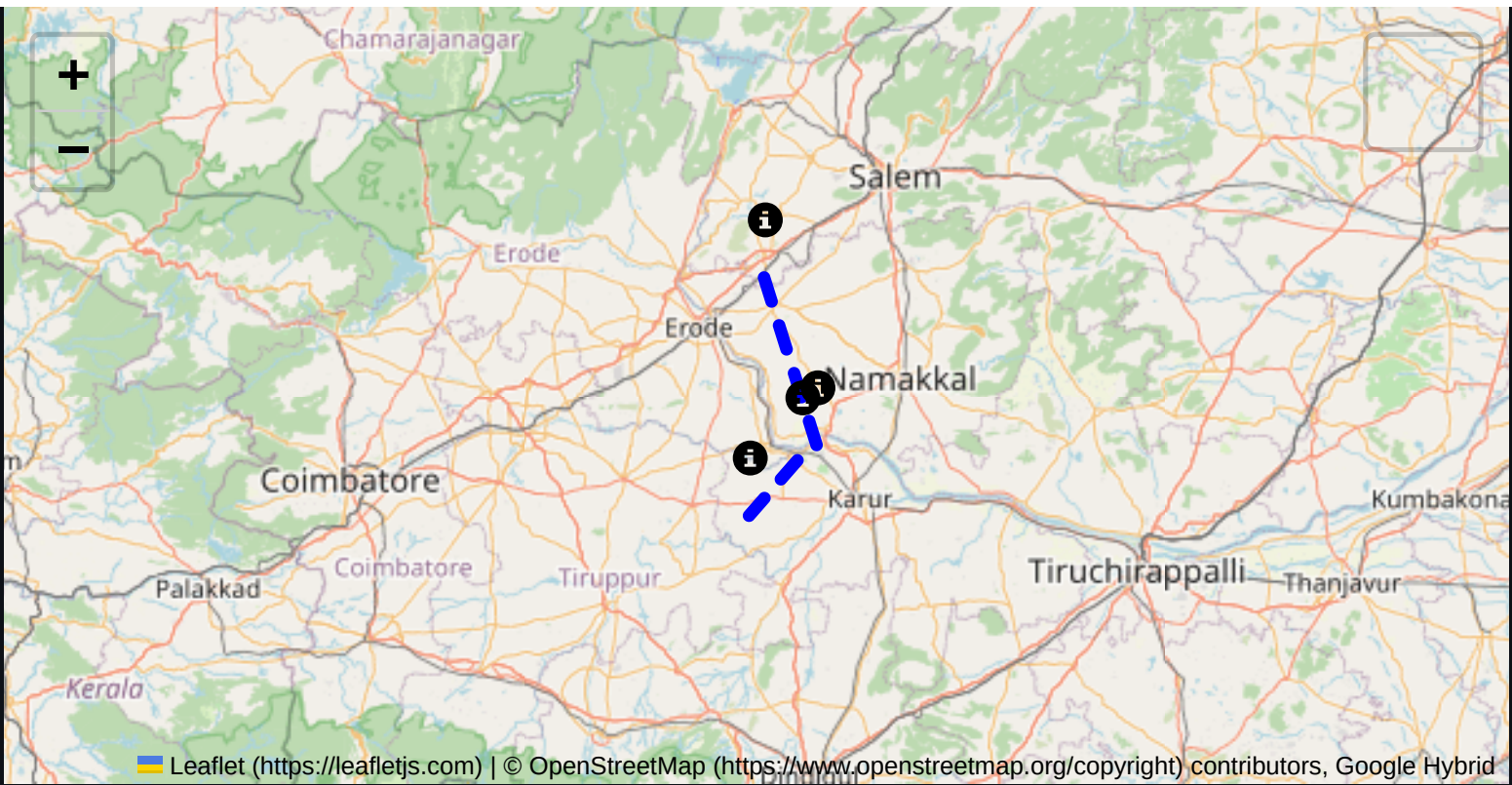
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

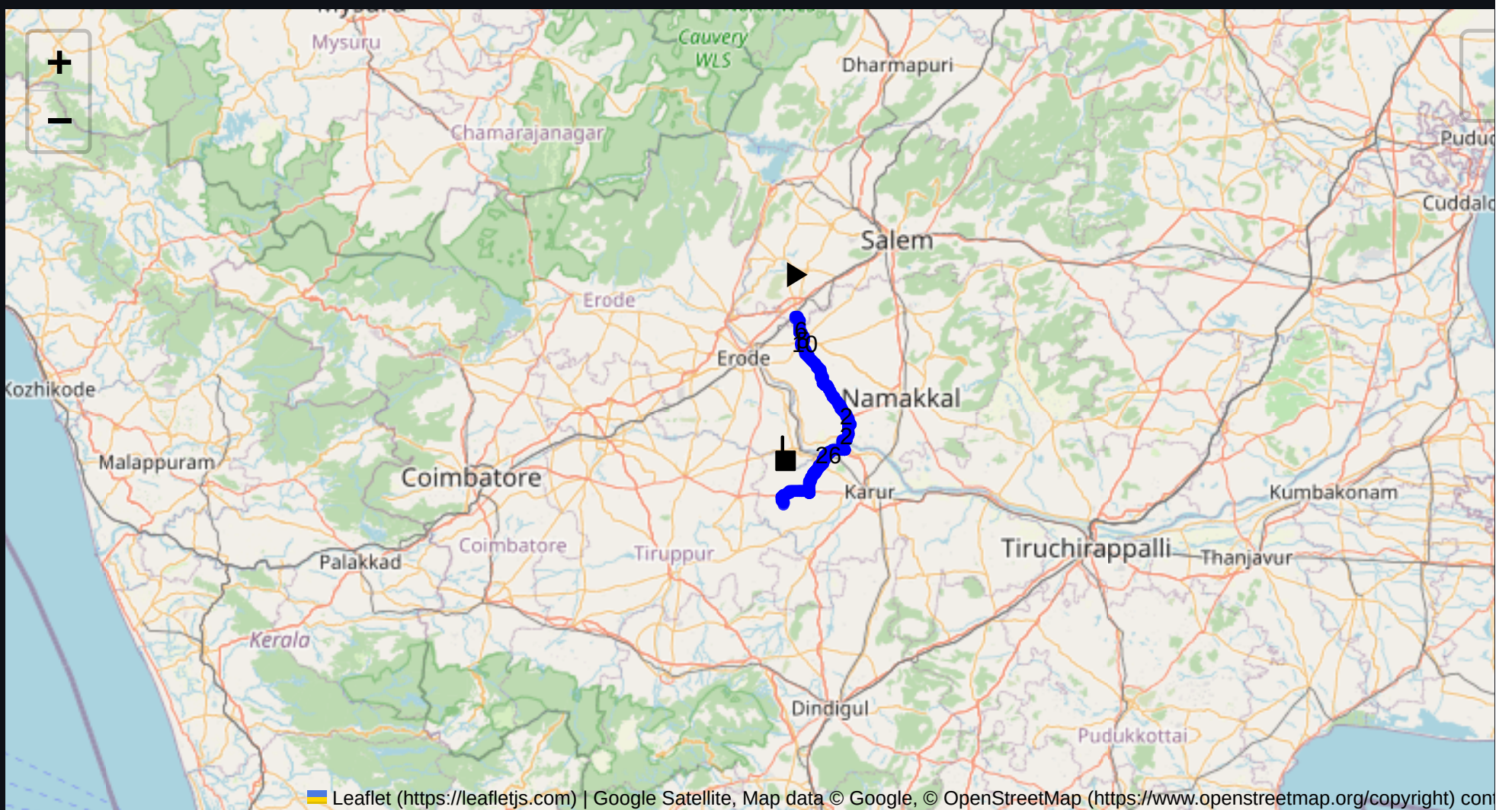
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 79.87 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (10.927388, 77.836965)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 79.87 kilometers and connects Sangagiri, Karaipalayam, Pongodai, and Thennilai South in Tamil Nadu, India. The journey travels primarily along state highways and rural roads. Starting from Sangagiri, the route moves eastward towards Karaipalayam, progresses through Pongodai, and ultimately reaches Thennilai South.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** The region experiences a tropical climate with hot summers, moderate monsoons, and mild winters.
- **Weather-Related Hazards:** During the monsoon season (June to September), heavy rains can lead to localized flooding, especially on rural and poorly maintained roads. Additionally, visibility may be reduced during heavy rainfall.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00 - 10:00 AM) and evening (5:00 - 7:00 PM) are typically peak hours with increased local traffic due to commercial and school activities.
- **Congestion-Prone Areas:** Town centers in Sangagiri and Karaipalayam can experience congestion, particularly during peak hours as local traffic mixes with through traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varied, with stretches of well-maintained state highways interspersed with narrower rural roads that may have uneven surfaces and limited markings.
- **Infrastructure:** Limited roadside amenities in rural sections; service stations and amenities are more available near larger towns like Sangagiri.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** If road closures or significant delays occur, consider bypassing Karaipalayam and Pongodai via NH81, reaching Thennilai South from a more southerly approach to avoid rural choke points and improve road quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Tamil Nadu follows India's national guidelines for the transport of hazardous materials, enforcing vehicle and safety standards. These include prohibition of materials beyond certain limits through populated areas and requiring detailed documentation.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Historical Incidents:** Scant public records of major incidents in this specific route. However, occasional reports of vehicle malfunctions and traffic mishaps exist due to road conditions and driver error.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** While passing through rural areas, drivers should be aware of local wildlife and agricultural activities. Minimize pollution and noise, as well as the risk of spills in these areas.

9. Analysis of Communication Coverage

- **Coverage:** Major towns and highways generally have good mobile network coverage. However, rural sections such as stretches between Karaipalayam and Pongodai may experience occasional dead zones.

10. Estimated Emergency Response Times

- **Emergency Response Times:** In urban areas like Sangagiri, emergency services can respond within 30 minutes. In rural parts of the route, response times may extend to over an hour due to distance and limited infrastructure.

11. Overall Summary of Risk Assessment

The route presents a moderate level of risk primarily due to potential weather disruptions and rural road conditions. Congestion in more populated areas can affect transport schedules. However, with proper planning, adherence to regulations, and awareness of these risks, safe transport of hazardous materials is achievable. Enhanced communication technologies and real-time weather updates are recommended to mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.07462, 78.00608	15 KM/Hr
2	Roundabout	High	11.07485, 78.00156	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16755, 78.01568	30 KM/Hr
9	Turn	Medium	11.15579, 78.02169	30 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14485, 78.02282	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit	
12	Turn	Medium	11.10084, 78.00555	30 KM/Hr	
13	Turn	High	11.07466, 78.00842	15 KM/Hr	
14	Turn	Medium	11.07578, 77.97518	30 KM/Hr	
15	Turn	High	11.05890, 77.95373	15 KM/Hr	
16	Turn	Medium	11.04837, 77.94890	30 KM/Hr	
17	Turn	High	11.04807, 77.94800	15 KM/Hr	
18	Turn	Medium	11.04673, 77.94766	30 KM/Hr	
19	Turn	High	11.04661, 77.94755	15 KM/Hr	
20	Turn	High	11.04677, 77.94601	15 KM/Hr	
21	Turn	Medium	11.04649, 77.94595	30 KM/Hr	
22	Turn	Medium	11.04190, 77.94682	30 KM/Hr	
23	Turn	Medium	11.03931, 77.94681	30 KM/Hr	
24	Turn	Medium	11.03753, 77.94595	30 KM/Hr	
25	Turn	Medium	11.03531, 77.94380	30 KM/Hr	
26	Turn	Medium	11.03443, 77.94256	30 KM/Hr	
27	Turn	High	11.03330, 77.93776	15 KM/Hr	
28	Turn	High	11.01499, 77.92643	15 KM/Hr	
29	Blind Spot	Blind Spot	11.01548, 77.92461	10 KM/Hr	
30	Turn	High	10.95908, 77.91206	15 KM/Hr	
31	Turn	High	10.95890, 77.91189	15 KM/Hr	
32	Turn	High	10.94917, 77.83270	15 KM/Hr	

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Velauthapalayam	11.0729521, 78.0023123	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07462, 78.00608



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07485, 78.00156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15579, 78.02169



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14485, 78.02282



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10084, 78.00555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.07466, 78.00842



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.07578, 77.97518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.05890, 77.95373



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.04837, 77.94890



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04807, 77.94800



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.04673, 77.94766



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04661, 77.94755



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04677, 77.94601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.04649, 77.94595



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.04190, 77.94682



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03931, 77.94681



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03753, 77.94595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03531, 77.94380



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.03443, 77.94256



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.03330, 77.93776



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.01499, 77.92643



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.01548, 77.92461



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95908, 77.91206



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95890, 77.91189



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.94917, 77.83270

